

BID CRUSADES

Official Evaluation & Judging Framework

Headstart · IIT Dharwad

Component	Evaluated By	Marks
Part A – Quantitative Business Evaluation	Organizers (Formula-Based)	50
Part B – Qualitative Strategic & Presentation	Judges	50

Final Score = Organizer Score (50) + Judge Score (50) = 100 Marks

This document details every formula, threshold, and scoring mechanism used in the competition. Organizer scores are fully formula-driven and locked before judge presentations. No manual intervention is permitted for Part A.

1. Part A – Organizer Quantitative Evaluation (50 Marks)

This component is fully formula-driven, computed immediately after disaster declaration, locked before judge presentations, and generated via the official scoring engine. No manual intervention is permitted.

1.1 Inputs Used by the System

- Team Name
- Asset ID List (from official asset sheet)
- Remaining Budget (■ Cr)
- Disaster Code (W / X / Y / Z)

1.2 Disaster Mechanism

After the auction concludes, one macroeconomic disaster is randomly triggered via the Disaster Roulette wheel. Each disaster has equal probability (25%) and impacts asset categories as follows:

Disaster	Cat A	Cat B	Cat C	Cat D
W	+20% ↑	0%	0%	−20% ↓
X	−20% ↓	+20% ↑	0%	0%
Y	0%	−20% ↓	+20% ↑	0%
Z	0%	0%	−20% ↓	+20% ↑

Each disaster: 1 category appreciates (+20%), 2 remain neutral (0%), and 1 depreciates (−20%). Post-disaster portfolio value is recalculated using: **Post Value** = $\sum (\text{Sale Price} \times \text{Multiplier})$

Multipliers: Appreciating = 1.20, Neutral = 1.00, Depreciating = 0.80

1.3 Business Strength (40 Marks)

A. Vertical Coverage Score (15 Marks)

Teams are evaluated on coverage of four hidden structural verticals. Each asset belongs to exactly one vertical. The verticals are:

- V1 – Core Value Creation
- V2 – Enablers & Technology
- V3 – Market Access & Distribution
- V4 – Stability & Strategic Protection

$$\text{Vertical Score} = (\text{Vertical Count} / 4) \times 15$$

Verticals Covered	Score
1	3.75
2	7.5
3	11.25
4	15.0

B. Risk Discipline Score (15 Marks)

Measures portfolio diversification across visible asset categories (A, B, C, D). The maximum category exposure is calculated as:

$$\text{Max Exposure} = \text{Highest Category Investment} / \text{Total Investment}$$

The 45% threshold is a benchmark. Teams at or below 45% receive full marks. Above 45%, the score decays aggressively using a **quadratic penalty** — the higher the concentration, the steeper the penalty:

$$\begin{aligned} \text{If Max Exposure} \leq 45\%: \text{Risk Score} &= 15 \text{ (full marks)} \\ \text{If Max Exposure} > 45\%: \text{Risk Score} &= 15 \times ((1 - \text{Max Exposure}) / 0.55)^2 \end{aligned}$$

Max Exposure	Risk Score	Penalty
≤ 45%	15.0	None (full marks)
50%	12.4	Moderate
55%	10.0	Significant
60%	7.9	Heavy
65%	6.1	Heavy
70%	4.5	Severe
75%	3.1	Severe
80%	2.0	Critical
90%	0.5	Near-zero
100%	0.0	Maximum penalty

Note: The quadratic decay means scores drop much faster than linear. A team at 60% exposure loses over half their risk marks. The strict binary threshold (≤45%) is reserved for the Bonus condition.

C. ROI Performance Score (10 Marks)

$$\text{ROI} = (\text{Post-Disaster Portfolio Value} - \text{Total Investment}) / \text{Total Investment}$$

Unlike vertical and risk scores, ROI uses a **continuous linear scale** rather than fixed tiers. This rewards every percentage point of return proportionally and ensures that teams taking calculated concentration risks can earn higher ROI marks to partially offset risk discipline penalties:

$$\begin{aligned} \text{ROI Score} &= \min(10, \max(0, \text{ROI} \times 50)) \\ \text{i.e. ROI Score} &= (\text{ROI} / 20\%) \times 10, \text{ clamped to } [0, 10] \end{aligned}$$

ROI	Score	Typical Strategy
+20%	10.0	All-in on appreciating category (max risk)
+15%	7.5	Heavy concentration, lucky disaster
+12%	6.0	~60% in appreciating category
+9%	4.5	Best case with ≤45% exposure (diversified)
+5%	2.5	Moderate diversification, favorable disaster

0%	0.0	Break even or balanced exposure
Negative	0.0	Net loss (depreciating category hit)

Business Strength = Vertical Score + Risk Score + ROI Score (Maximum: 40 Marks)

1.4 Bonus Score (Maximum 10 Marks)

Three independent binary conditions are evaluated. These are intentionally **hard thresholds** — each condition is either met or not. Unlike the dynamic Risk Discipline score, bonus conditions reward only strict discipline:

Condition 1: Portfolio contains assets across all 4 verticals AND at least one additional asset (minimum 5 assets total).

Condition 2: $ROI \geq 8\%$.

Condition 3: Maximum Category Exposure $\leq 45\%$.

Conditions Met	Bonus Marks
0	0
1	3
2	6
3	10

Organizer Score = Business Strength (40) + Bonus (10) (Maximum: 50 Marks)

1.5 Implementation Note

All Part A calculations are executed by the official scoring engine. The system:

- Applies disaster multipliers to recalculate post-disaster portfolio values
- Computes category exposures and ROI
- Evaluates vertical presence across hidden verticals
- Calculates business strength using dynamic risk scoring
- Evaluates bonus conditions with strict binary thresholds
- Generates team summary PDFs with full score breakdowns
- Records organizer scores — scores are final and non-negotiable

2. Part B – Judge Evaluation (50 Marks)

This component is evaluated by the judging panel during team presentations. Judges assess qualitative strategic reasoning and presentation quality.

2.1 Strategic Coherence & Market Logic (25 Marks)

A. Strategic Depth & Structural Logic (15 Marks)

Judges evaluate:

- Internal coherence of asset selection
- Logical business structuring
- Scalability reasoning
- Alignment between portfolio and business thesis

B. Market Feasibility & Positioning (10 Marks)

Judges evaluate:

- Sector viability
- Competitive positioning
- Revenue logic
- Market defensibility

2.2 Presentation & Communication (25 Marks)

A. Clarity & Structure (10 Marks)

- Structured argument flow
- Clear articulation of thesis
- Logical sequencing

B. Data Use & Disaster Explanation (10 Marks)

- Accurate explanation of disaster impact on portfolio
- Correct ROI interpretation
- Analytical depth in crisis handling

C. Q&A Defense (5 Marks)

- Composure under questioning
- Logical rebuttal
- Numerical awareness
- Strategic consistency

Judge Total = Strategic Depth + Market Feasibility + Clarity + Data Use + Q&A (Maximum: 50 Marks)

3. Judge Scoring Sheet

Use this sheet to record scores for each team. One sheet per team.

Team Name:

Judge Name:

#	Criteria	Max Marks	Score
	STRATEGIC COHERENCE & MARKET LOGIC	25	
A	Strategic Depth & Structural Logic	15	
B	Market Feasibility & Positioning	10	
	PRESENTATION & COMMUNICATION	25	
C	Clarity & Structure	10	
D	Data Use & Disaster Explanation	10	
E	Q&A Defense	5	
	JUDGE TOTAL	50	

Comments / Notes:

4. Final Score Calculation

Final Total = Organizer Total (50) + Judge Total (50) (Maximum: 100 Marks)

Component	Sub-component	Max Marks
Part A: Organizer Score	A. Vertical Coverage	15
	B. Risk Discipline (Dynamic)	15
	C. ROI Performance	10
	Bonus (3 conditions)	10
	Subtotal	50
Part B: Judge Score	Strategic Depth & Logic	15
	Market Feasibility	10
	Clarity & Structure	10
	Data Use & Disaster Explanation	10
	Q&A Defense	5
	Subtotal	50
FINAL TOTAL		100

5. Integrity Safeguards

- Organizer scores are formula-driven and generated before presentations.
- Judges only input qualitative marks (Part B).
- All totals are automatically computed by the scoring engine.
- No manual score editing is permitted for Part A.
- Team summary PDFs are auto-generated with full score breakdowns.
- The dynamic risk scoring formula is transparent — teams can verify their own scores.

6. Quick Formula Reference

Formula	Expression
Total Investment	100 – Remaining Budget
Post-Disaster Value	Σ (Sale Price $\times$ Disaster Multiplier)
ROI	(Post-Disaster Value – Total Investment) / Total Investment
Max Exposure	Highest Category Investment / Total Investment

Vertical Score (15)	$(\text{Verticals Covered} / 4) \times 15$
Risk Score (15)	$\leq 45\%: 15 \mid > 45\%: 15 \times ((1 - \text{Exp}) / 0.55)^2$
ROI Score (10)	$\min(10, \max(0, \text{ROI} \times 50))$ — linear continuous
Business Strength (40)	Vertical + Risk + ROI Score
Bonus (10)	0 cond:0 1:3 2:6 3:10
Organizer Total (50)	Business Strength + Bonus
Final Total (100)	Organizer Total + Judge Total